Namma Space — Round 1 technical abstract

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1. What was built

The deliverable is a free-roaming web viewer of one classroom. The picture is a Gaussian splat: the iPhone photos were trained into about one million coloured points, so the walls, desks, and boards keep the real photograph. Walking does not use those points as colliders. A measured floor plan, taken from the same capture's LiDAR, stops the visitor at the walls and at one closed door. A plan in the corner of the screen shows the room, that door, and where the visitor is standing.

2. Dataset

The room is the MuSHRoom classroom long capture: an iPhone 12 Pro Max, November 2023, released CC BY 4.0. It has 797 posed colour frames and a matching depth image for each frame. Colour used for training is 738 by 994. Depth is the phone LiDAR. The capture is a public indoor room, used because a same-day phone test on this desk contained one frame only, which cannot train a room. The photographs and the depth remain the dataset authors' work.

3. Standard operating procedure

The capture procedure this pipeline expects, and the one a later home scan should follow:

- Record on the 1× camera of a LiDAR iPhone, in Record3D (EXR + JPG) or an equivalent package that stores a photo, a depth image, a confidence map, and a pose for every frame.
- Hold the phone portrait, about chest height, tilted slightly down so each frame sees floor and wall. Do not roll it to landscape. A sideways hold turns the floor into a wall and the plan follows.
- Walk a slow loop of the room, then across the middle, then through the doorway without stopping the recording. Pause at corners. Keep neighbouring views overlapping. LiDAR reaches about 5 m, so walk up to each wall.
- Keep people out of the shot. Lock the exposure so a door jamb is sharp when you pause. A gimbal is useful only if it steadies that walk. A spin from one spot is not a scan.
- One continuous take keeps one coordinate frame. If tracking drops, start again with the same doorway in both recordings.
- Before leaving, check that depth, pose, and colour exist for the start, the middle, and the end, and that the doorway is seen from both sides.

4. Pipeline

- Import. Each frame becomes a colour image, a depth image, lens numbers, and a camera-to-world pose. The eight ways of attaching the picture to the pose are scored. A winner is kept only when it is clearly better, or when a later geometry check finds a floor, a room 2–4 m tall, and the cameras inside it.
- Up. The floor is levelled from the camera motion and the depth, then snapped to a vertical axis. The plan is drawn with north as the negative z axis of that frame.
- Train. 550 of the sharpest frames, people frames removed, seed the points from the LiDAR at a 4 cm voxel. Colour is trained for 30,000 steps. Depth is a light constraint, weight 0.05, so the scale stays in metres. The config is pipeline/mushroom_classroom_1m.json.
- Walls. A separate measured layout, from the same depth, supplies the floor polygon, the wall segments, and the closed door. Furniture in the photograph is not a collider.

- Viewer. The splat is drawn with Spark and Three.js. WASD moves a body of radius 0.18 m. The eye is 1.60 m above the measured floor. A move that would enter a wall or the closed door is refused.

5. Check

24 views held out of training, protocol raw pose, no test-time alignment: mean PSNR 22.9 dB, mean SSIM 0.739, depth mean absolute error 7.0 cm against the LiDAR (median 2.7 cm). The export contains 999,454 Gaussians. The spawn point used by the viewer sits inside the floor polygon and at least 0.18 m from every wall.

6. What this is not

An iPad Pro house scan (ARKitScenes 45663064, 256×192, held sideways) was started on the measured-wall pipeline and then stopped. Its camera roll was not confident, the plan came out turned 90 degrees, and the colour stream is too small to match this classroom. That run is not part of the submission. A one-frame LiDAR photo taken on 1 November 2026 confirmed that the phone can save depth, confidence, and a pose. It is not a room.

7. How to view and how to rebuild

The web viewer is the GitHub Pages site named in Proof-of-Concept.pdf. Walking instructions are on the page. RECONSTRUCT.md in the repository gives the train command and the sample directory test/sample, which holds eight frames in the same JSON shape as the full capture. The trained scene.ply is about 210 MB and is attached to the repository release, because GitHub rejects a single file of that size in the git history.